

Mini Report: Refactoring CustomerManager

This report briefly describes how I refactored the CustomerManager class.

First, we changed some variable names like x, y, z to more meaningful ones like customer_name, purchases, etc., so the code is easier to read.

We also moved the tax rate and thresholds into the constructor instead of keeping them hard-coded. That way, they can be adjusted if needed.

To reduce code duplication, we reused add_customer in both add_purchase and add_purchases, so they just call the same function with different inputs.

For the report generation, we separated out the logic to calculate total with tax and classify the customer into separate methods (like classify_customer), which makes generate_report cleaner.

Shipping fee logic was a bit repetitive, so we made sure CustomerManager only checks for heavy items (weight > 20), and the fragile item check remains in the external function calculate_shipping_fee_for_fragile_items to keep the test working.

After these changes, the code passed all the tests provided.

This refactoring helps improve readability and makes the class more flexible for future changes.